# Parallel and distributed Computing

## Lecture 31

Bazam-e-Jahan

Lecturer

The University of Faisalabad

# Lecture 31
# A Decentralized Algorithm

# Characteristics of Decentralized Algo

- no machine has complete information about the system state.
- Machine make decision based only on local information
- Failure of one machine does not ruin the algorithm
- Three is no implicit assumption that a global clock exists

# Decentralized Algorithm

- Based on the distributed hash table DHT system structure previously introduced
  - Peer-to-peer
  - object names are hashed to find the successor node that will store them
- Here, we assume that n replicas of each object are stored

# Decentralized Algorithm

- Every replica has a coordinator that controls access to it (the coordinator is the node that stores it)
- For a process to use the resource it must receive permission from m › n/2 coordinators
- This guarantees exclusive access as long as a coordinator only grants access to one process at a time

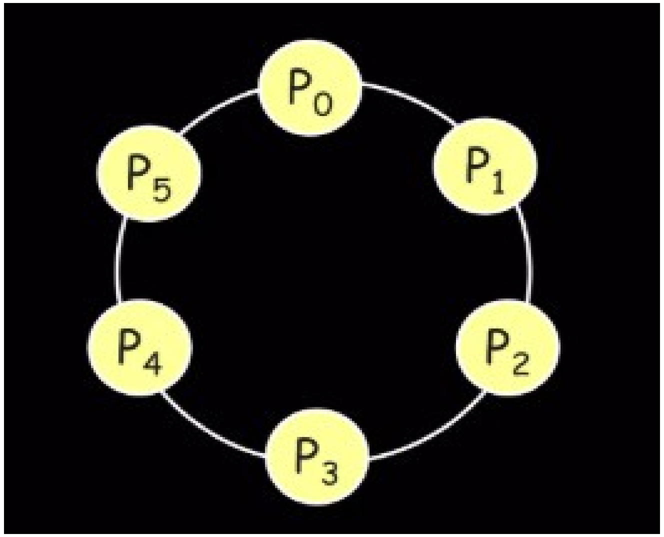
# Decentralized Algorithm

- The coordinator notifies the requester when it has been denied access as well as when it is granted.
    - Requester must "count the votes", and decide whether or not overall permission has been granted or denied
- If a process (requester) gets fewer than m votes it will wait for a random time and then ask again

# Lecture 32
# A Token Ring Algorithm

# Token Ring algorithm

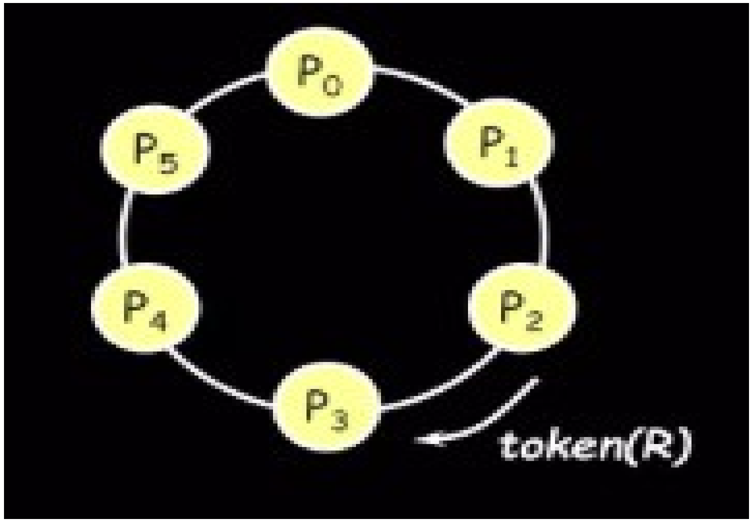
- Assume known group of processes
  - Some ordering can be imposed on group
  - Construct logical ring in software
  - Process communicates with neighbor



# Token Ring algorithm



- Initialization
  - Process O gets token for resource R
- Token circulates around ring
  - From P; to P (i+1)mod N
- When process acquires token
  - Checks to see if it needs to enter critical section
  - If no, send token to neighbor
  - If yes, access resource
    - Hold token until done



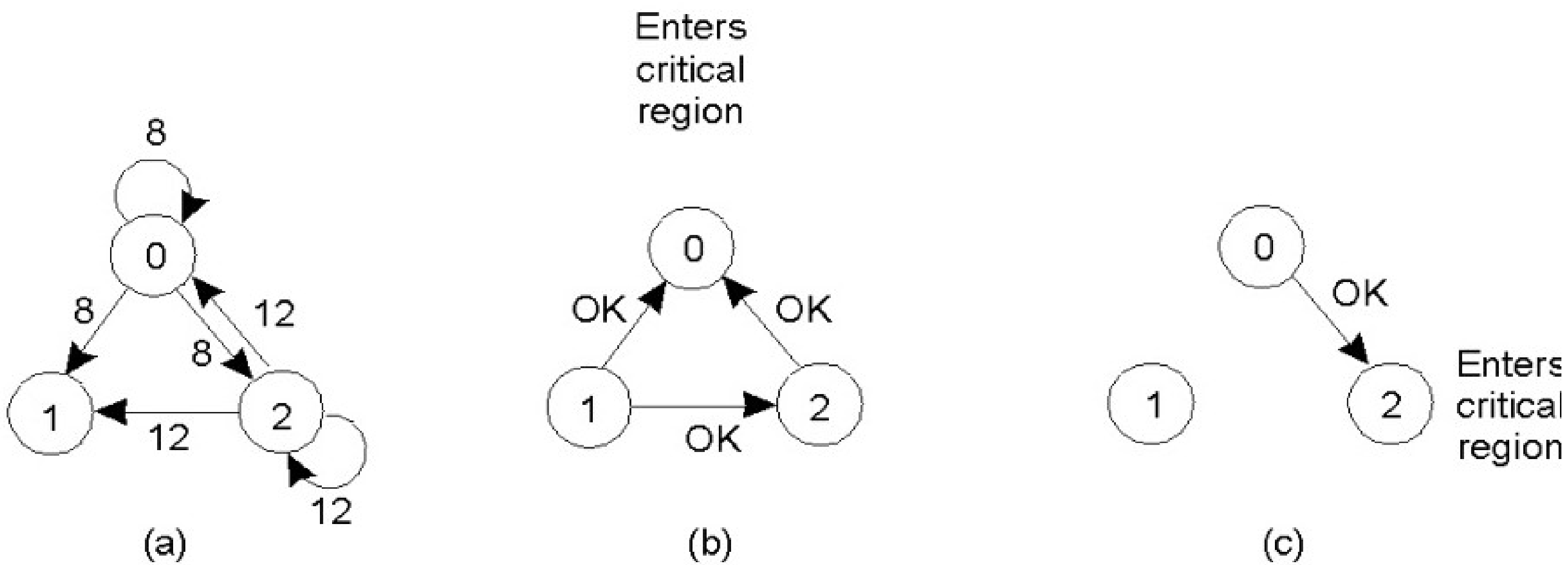
# Token Ring algorithm



- Only one process at a time has token
  - Mutual exclusion guaranteed
- Order well-defined
  - Starvation cannot occur
- If token is lost (e.g. process died)
  - It will have to be regenerated
- Does not guarantee FIFO order

# Lecture 33
# A Distributed Algorithm

# Distributed algorithm




a) Two processes want to enter the same critical region at the same moment.
b) Process 0 has the lowest timestamp, so it wins.
c) When process 0 is done, it sends an OK also, so 2 can now enter the critical region.

# Lecture 34 Comparison of these for Algorithm

# Read it by yourself

# Comparison Analysis

## Mutual Exclusion Algorithms Comparison

| Criteria | Centralized | Distributed | Decentralized | Token Ring |
|---|---|---|---|---|
| Control Mechanism | Single coordinator (central server) | All nodes communicate to reach consensus | Multiple coordinators | Logical ring passes a token |
| Request Message Complexity | O(1) (request to coordinator) | O(n) (each node sends to all others) | O(1) to O(n), depending on load balancing | O(1) (only token passing) |
| Synchronization Delay | Low | Medium to high | Medium | Low to medium |
| Bottleneck | Yes (coordinator) | No | No | No |
| Single Point of Failure | Yes | No | Reduced (due to redundancy) | Yes (token loss or node failure) |
| Fairness | Depends on queueing at coordinator | Generally fair (logical timestamps) | Can be unfair depending on design | Fair (token rotates through all nodes) |
| Fault Tolerance | Low (if coordinator crashes) | Higher (can tolerate failures) | Medium to high (redundancy needed) | Medium (token regeneration required) |
| Message Overhead | 2 messages per CS entry | 2(n-1) messages per CS entry | Variable (depends on architecture) | 1 message per CS entry (token pass) |